

EC-Council



Digital Forensics Essentials

MODULE 4

Data Acquisition and Duplication / Defeating Anti-Forensic Techniques



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Data Acquisition and Duplication

SUBTOPIC 1

Module Objectives

- ➔ Understanding the Data Acquisition Fundamentals
- ➔ Understanding the Different Types of Data Acquisition
- ➔ Understanding the Data Acquisition Format
- ➔ Understanding the Data Acquisition Methodology



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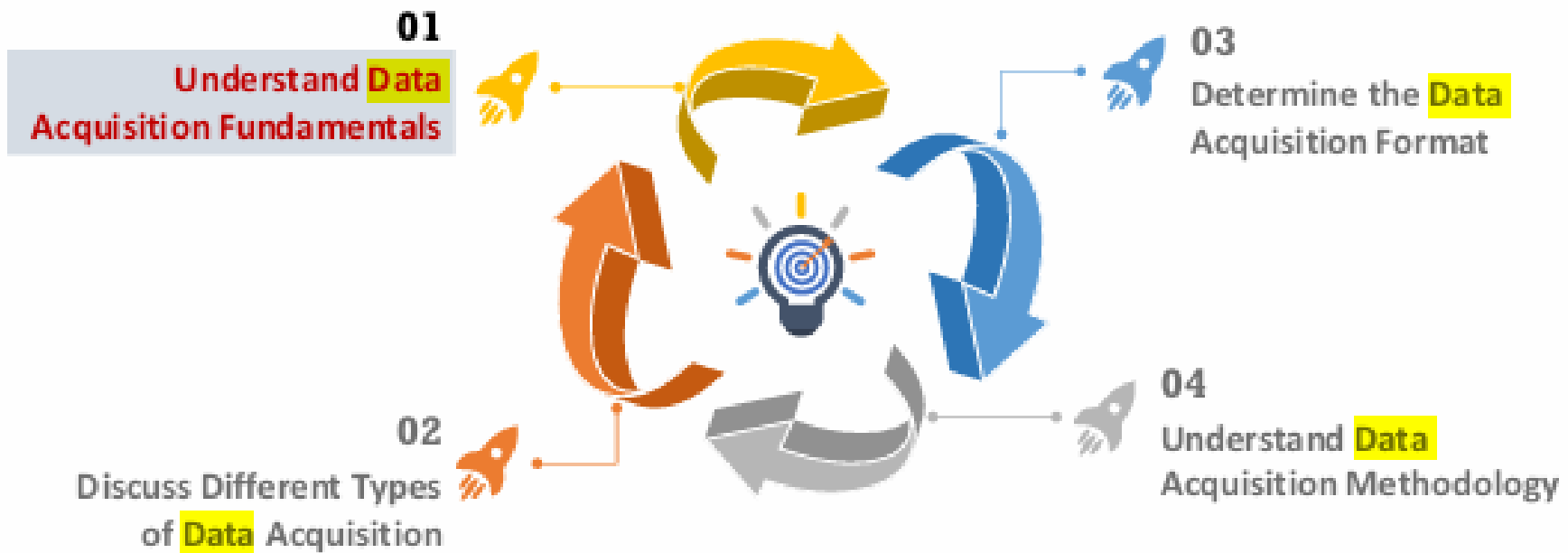
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Module Flow



Data Acquisition



- ❑ **Data** acquisition is the use of established methods to **extract Electronically Stored Information (ESI)** from suspect computer or storage media to gain insight into a crime or an incident
- ❑ Investigators must be able to verify the accuracy of acquired **data**, and the complete **process should be auditable and acceptable in the court**

Data Acquisition Categories



Live Acquisition

It involves collecting **data** from a system that is powered **ON**



Dead Acquisition (Static Acquisition)

It involves collecting **data** from a system that is powered **OFF**



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Live Acquisition



- ❑ Live **data** acquisition involves collecting volatile **data** from a live system
- ❑ Volatile information assists in determining the **logical timeline** of the security incident, and the possible users responsible
- ❑ Live acquisition can then be **followed by static/dead acquisition**, where an investigator shuts down the suspect machine, removes the hard disk and then acquires its forensic image

Types of **data** captured during live acquisition

System **Data**

- Current configuration
- Running state
- Date and time
- Current system uptime
- Running processes
- Logged on users
- DLLs or shared libraries
- Swap files and temp files

Network **Data**

- Routing tables
- ARP cache
- Network configuration
- Network connections

Order of Volatility

- ❑ When collecting evidence, an investigator needs to evaluate the **order of volatility** of data depending on the suspect machine and the situation

According to the RFC 3227, below is an example of the order of volatility for a typical system:

- 01 Registers and cache
- 02 Routing table, process table, kernel statistics, and memory
- 03 Temporary system files
- 04 Disk or other storage media
- 05 Remote logging and monitoring data that is relevant to the system in question
- 06 Physical configuration, and network topology
- 07 Archival media

<https://toolsietf.org>



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Dead Acquisition



01

Dead acquisition is defined as the acquisition of data from a suspect machine that is **powered off**

02

Dead acquisition usually involves acquiring data from storage devices such **hard drives, DVD-ROMs, USB drives, flash cards,** and **smart phones**

03

Examples of static data: emails, word documents, web activity, spreadsheets, slack space, unallocated drive space, and various deleted files



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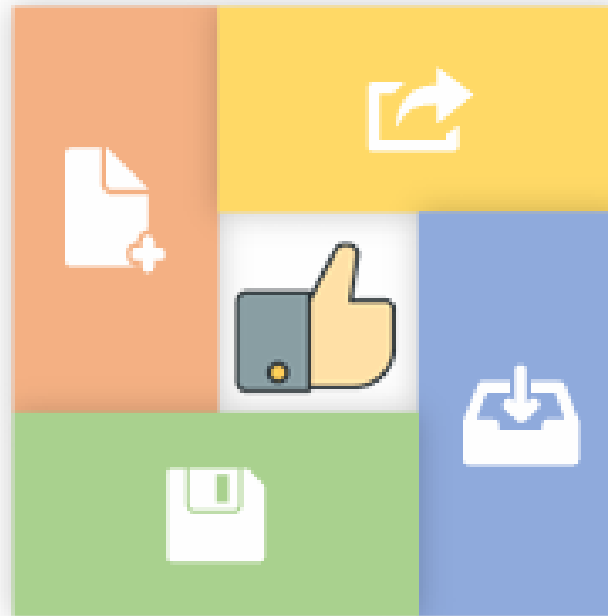
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Rules of Thumb for Data Acquisition

- ❑ Do not work on original digital evidence. Create a bit-stream/logical image of a suspicious drive/file to work on.

- ❑ Use clean media to store the copies



- ❑ Produce two or more copies of the original media
 - The first is the **working copy** to be used for analysis
 - The other copies act as the **library/control copies** that are stored for **disclosure** purposes or in the event that the working copy gets corrupt
- ❑ Upon creating copies of original media, verify the **integrity of copies** with the original

Module Flow



Types of Data Acquisition

Logical Acquisition

- ❑ Logical acquisition allows an investigator to capture only selected **files** or **files types** of interest to the case
- ❑ Examples of logical acquisition include:
 - Email investigation that requires collection of **Outlook .pst or .ost files**
 - Collecting specific records from a large **RAID server**

Sparse Acquisition

- ❑ Sparse acquisition is similar to logical acquisition, which in addition **collects fragments of unallocated data, allowing investigators to acquire deleted files**
- ❑ Use this method when inspection of the entire drive is not required



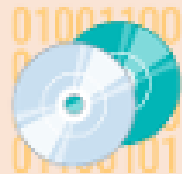
Types of Data Acquisition (Cont'd)

Bit-Stream Imaging

Bit-stream imaging creates a **bit-by-bit copy** of a suspect drive, which is a cloned copy of the entire drive including all its sectors and clusters, which allows forensic investigators to **retrieve deleted files or folders**

Bit-stream disk-to-image file

- ❑ It is the most common method used by **forensic investigators**
- ❑ The created image file is a bit-by-bit replica of the suspect drive
- ❑ Tools used: ProDiscover, EnCase, FTK, The Sleuth Kit, X-Ways Forensics, etc.



Bit-stream disk-to-disk

- ❑ Disk-to-image copying is not possible in situations where
 - The suspect drive is very old and incompatible with the imaging software
 - Investigator needs to recover credentials used for websites and user accounts
- ❑ To overcome this situation, investigators can create a **disk-to-disk bit-stream** copy of the target media
- ❑ While creating a disk-to-disk copy, investigators can adjust the **target disk's geometry** (its head, cylinder, and track configuration) to align with the suspect drive. This results in smooth data acquisition process.
- ❑ Tools used: Encase, Tableau Forensic Imager, etc.

Module Flow



Determine the Data Acquisition Format

Raw Format

- ❑ Raw format creates a bit-by-bit copy of the suspect drive. Images in this format were are usually obtained by using the **dd** command.

Advantages

- **Fast** data transfers
- Minor data read errors on **source drive** are ignored
- Read by most of the forensic tools

Disadvantages

- Requires same amount of **storage** as that of the original media
- Tools (mostly open source) might fail to recognize/collect **marginal** (bad) **sectors** from the suspect drive



Determine the Data Acquisition Format (Cont'd)

Proprietary Format

- ❑ **Commercial forensics tools** acquire data from the suspect drive and save the image files in their own formats

They offer certain features which include the following:

- Option to compress the image files of the evidence disk/drive in order to **save space on the target media**
- Ability to **split an image** into multiple **segments**, in order to save them to smaller target media such as CD/DVD, while maintaining their integrity
- Ability to **incorporate metadata** into the image file, which includes date and time of acquisition, hash values of the files, case details, etc.

Disadvantages

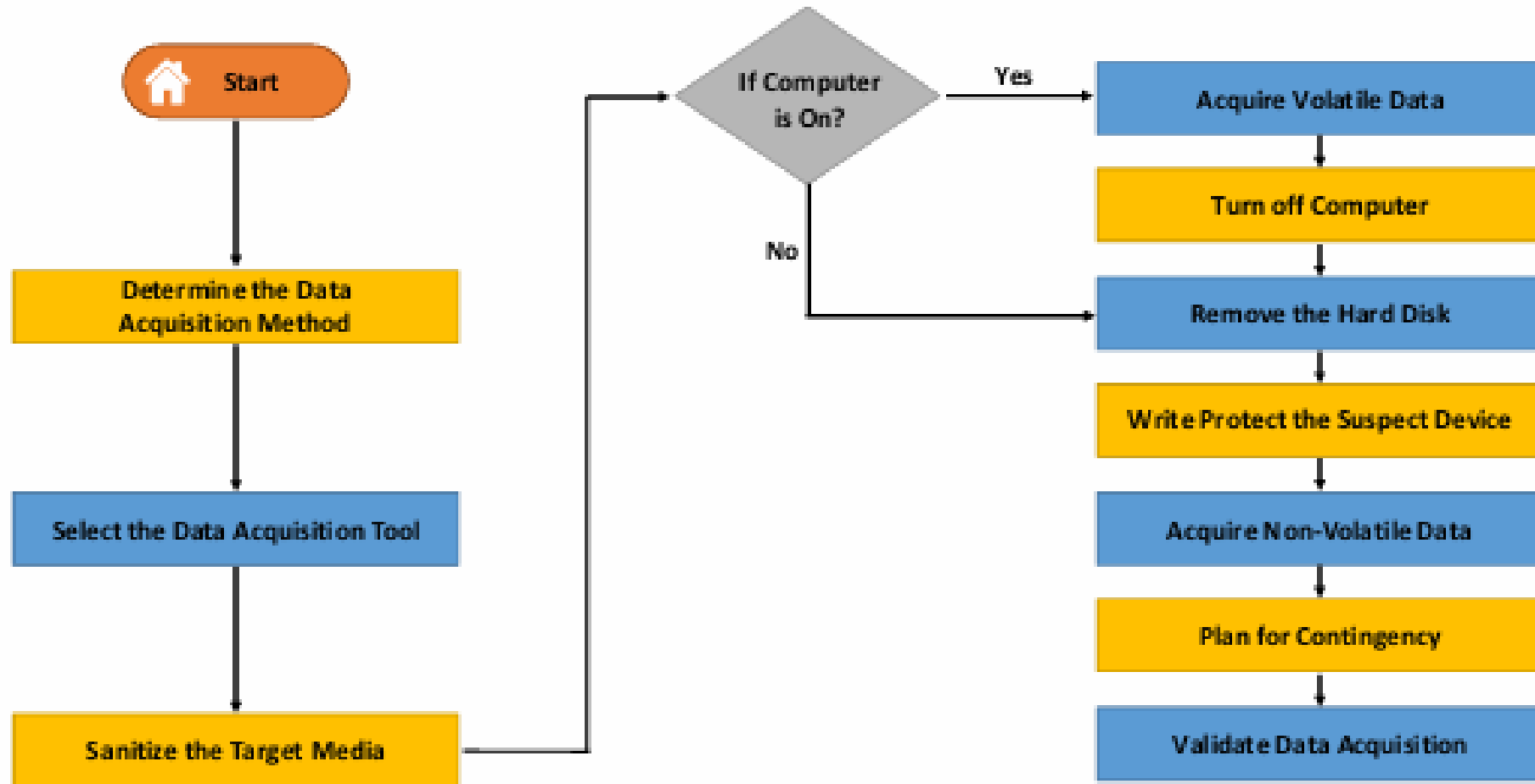
- Image file format created by one tool may not be supported by other tool(s)



Module Flow



Data Acquisition Methodology



Step 1: Determine the Best Data Acquisition Method

☐ An investigator needs to identify the **best data acquisition method** suitable for the investigation, depending on the situation the investigator is presented with

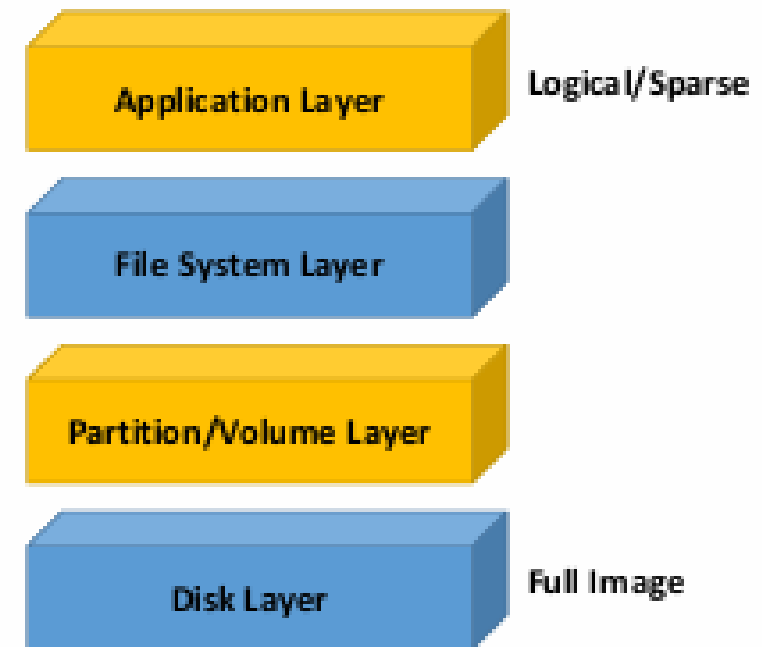
☐ These situations include:

- Size of the suspect drive
- Time required to acquire the image
- Whether the investigator can retain the suspect drive

☐ **Example:**

- In case the original evidence drive needs to be returned to the owner, as in the case of a discovery demand for a civil litigation case, check with the requester (lawyer or supervisor) whether logical acquisition of the disk is acceptable. If not, you may have to go back to the requester.

☐ Investigators need to acquire only the **data that is intended** to be acquired



Step 2: Select the Data Acquisition Tool

Mandatory Requirements

2 ☐ Investigators need to choose the **right tool for data acquisition** based on the type of acquisition technique they choose. When it comes to imaging tools, they need to choose the tools that satisfy certain requirements.

- 01 The tool should not change the **original content**
- 02 The tool should **log I/O errors** in an accessible and readable form, including the type of the error and location of the error
- 03 The tool must have the ability to pass **scientific** and **peer** review. Results must be repeatable and verifiable by a third party if necessary.
- 04 The tool should **alert the user** if the source is larger than the destination
- 05 The tool should create a **bit-stream copy** of the original content when there are no errors in accessing the source media
- 06 The tool should create a **qualified bit-stream copy** (a qualified bit-stream copy is defined as a duplicate except in identified areas of the bit-stream) when I/O errors occur while accessing the source media

Step 3: Sanitize the Target Media

- ❑ Investigators must properly **sanitize** the target media in order to any prior data residing on it, before it is used for collecting forensic data
- ❑ Post investigation, they must dispose this media by following the same standards, so as to **mitigate the risk** of unauthorized disclosure of information, and ensure its confidentiality
- ❑ The following are some standards for sanitizing media:
 - Russian Standard, GOST P50739-95
 - German: VSITR
 - American: NAVSO P-5239-26 (MFM)
 - American: DoD 5220.22-M
 - American: NAVSO P-5239-26 (RLI)
 - NIST SP 800-88



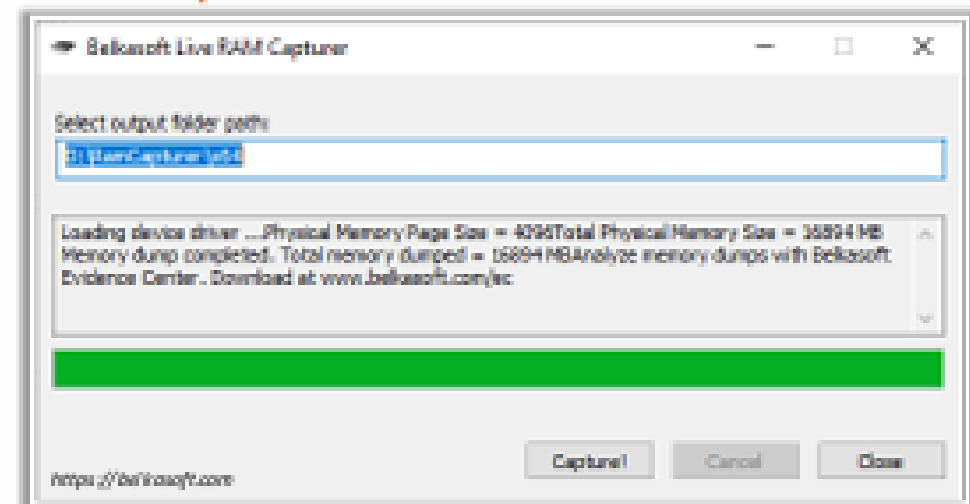
Step 4: Acquire Volatile Data

- ☐ Volatile data acquisition involves collecting data that is **lost** when the computer is shut down or restarted
- ☐ This data usually corresponds to running processes, logged on users, registries, DLLs, clipboard data, open files, etc.

Acquire Volatile Data from a Windows Machine

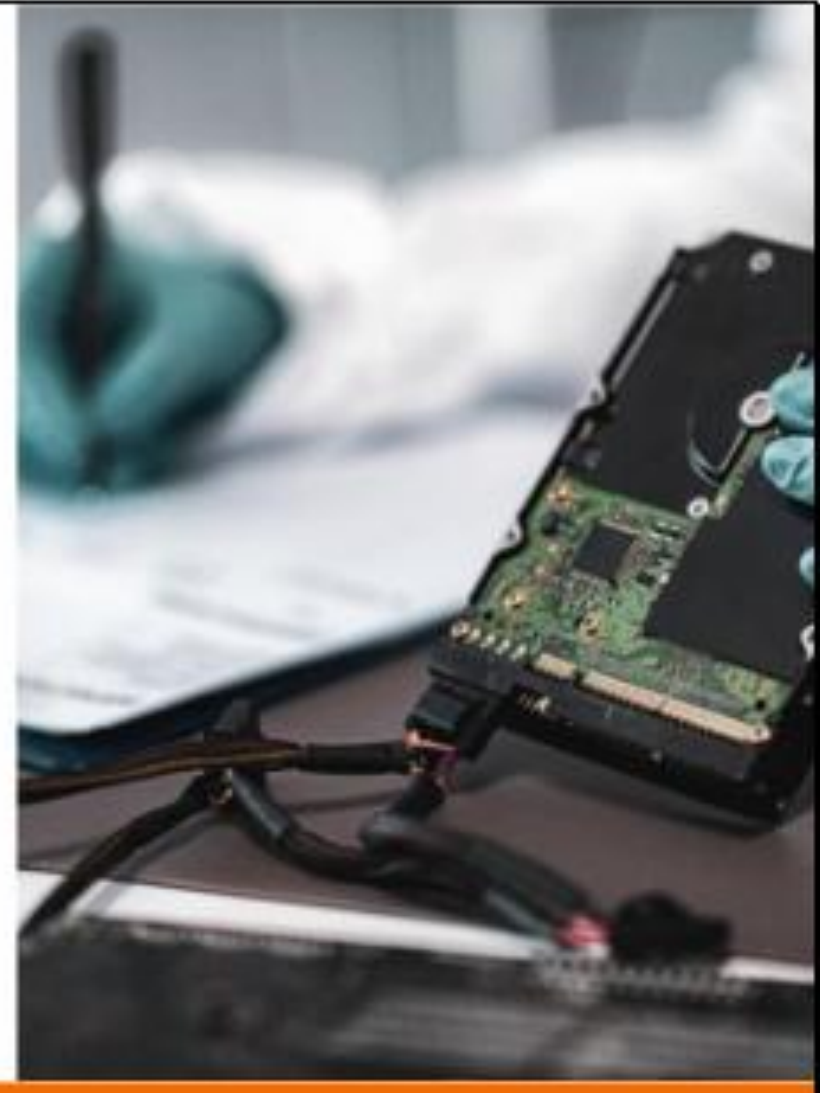
- ☐ Belkasoft Live RAM Capturer is a forensic tool that allows **extracting** the entire contents of a computer's **volatile memory**
- ☐ It saves the image files in .mem format

Note: While performing live acquisition, an investigator must be aware of the fact that working on a live system may alter the contents of RAM or processes running on the system. Any involuntary action performed on the system may potentially make the system inaccessible.



Step 5: Enable Write Protection on the Evidence Media

- ❑ It is necessary to write protect the suspect drive using write blockers to **preserve and protect the evidence** contained in it
- ❑ A write blocker is a hardware device or software application that allows data acquisition from the storage media without altering its contents
- ❑ It blocks write commands, thus allowing **read-only access** to the storage media
 - If hardware write blocker is used:
 - Install a write blocker device
 - Boot the system with the examiner-controlled operating system
 - Examples of hardware devices: CRU® WiebeTech® USB WriteBlocker , Tableau Forensic Bridges, etc.
 - If software write blocker is used:
 - Boot the system with the examiner-controlled operating system
 - Activate write protection
 - Examples of software applications: SAFE Block, MacForensicsLab Write Controller, etc.





01

Step 6: Acquire Non-volatile Data

Non-volatile data can be acquired in both live acquisition and dead acquisition. It mainly involves **acquiring data from a hard disk**.

02

There is no significant difference in the amount of data acquired from a hard disk between the live and dead acquisition methods

03

Live Acquisition of a hard disk is performed by using remote acquisition tools (e.g. netcat), and bootable CDs or USBs (e.g. CAINE); while dead acquisition involves removing the hard disk from the suspect drive, connecting it to a forensic workstation, write-blocking the hard disk, and running a forensic acquisition tool on the disk



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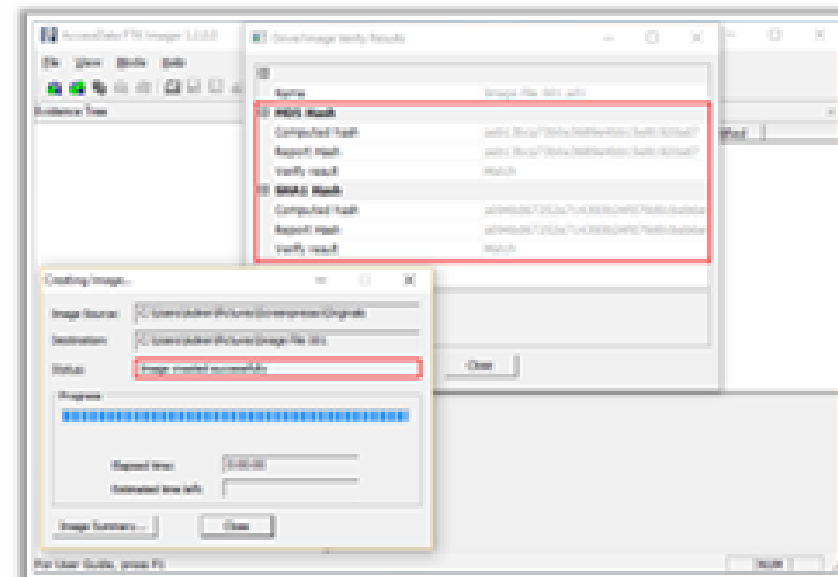
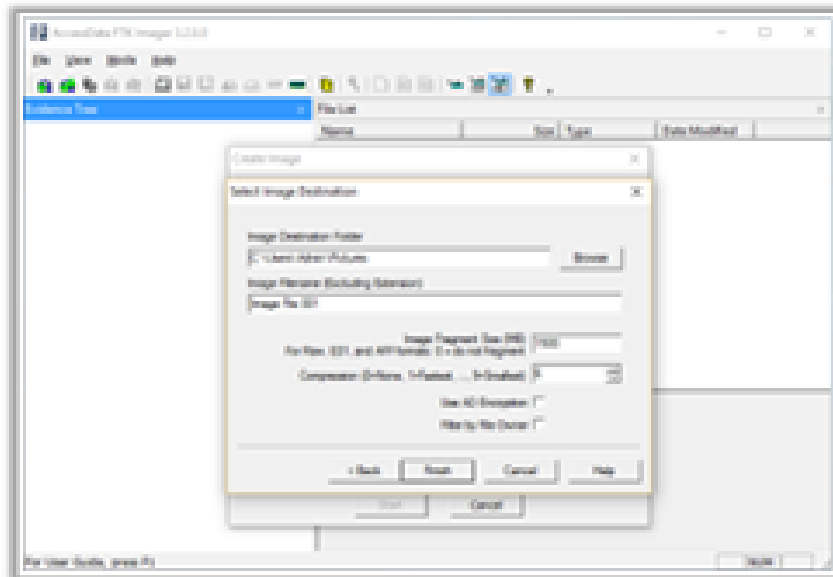


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Step 6: Acquire Non-volatile Data (Using a Windows Forensic Workstation)

- ❑ To acquire forensic image of a hard disk during **dead acquisition**, remove the hard disk, connect it to a forensic workstation, enable write-blocker, and run a forensic imaging tool (e.g. AccessData FTK Imager) on the workstation
- ❑ AccessData FTK Imager is a **disk imaging program** which can preview recoverable data from a disk of any kind and also **create copies**, called forensics images, of that data



<http://acris.idaho.com>

Step 7: Plan for Contingency

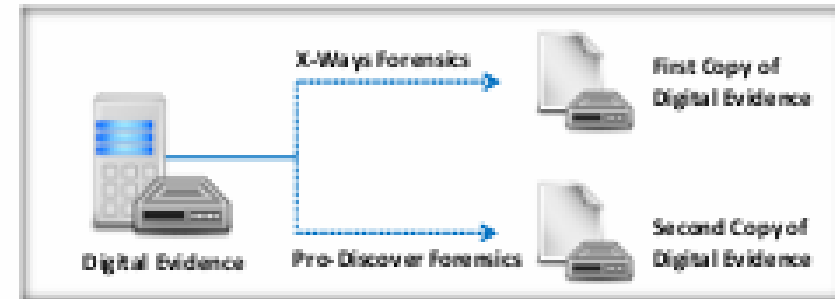
- Investigators must prepare for contingencies such as when the **hardware or software does not work**, or a failure occurs during acquisition

Hard Disk Data Acquisition

Investigators must create at least **two images of the digital evidence** collected, in order to preserve it. If one copy of the digital evidence recovered becomes corrupt, investigators can then use the other copy.

Imaging Tools

If you possess more than one **imaging tool**, such as Pro-DiscoverForensics or AccessData FTK Imager, it is recommended to create the **first image with one tool** and the **second image with the other tool**. If you possess only one tool, make two or more images of the drive using the same tool.



Step 7: Plan for Contingency (Cont'd)

Hardware Acquisition Tool



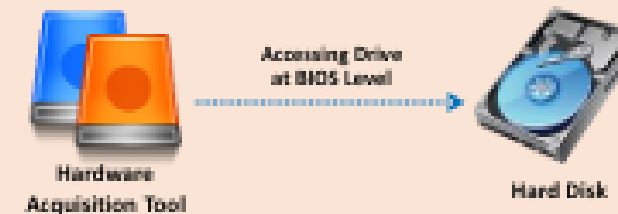
Consider using a hardware acquisition tool (such as **UFED Ultimate** or **IM SOLO-4 G3 IT RUGGEDIZED**) that can **access the drive at the BIOS level** to copy data in the Host Protected Area (HPA)



Drive Decryption



Be prepared to deal with encrypted drives that need the user to provide the **decryption key** for decrypting. Microsoft includes a full disk encryption feature (BitLocker) with select editions of Windows Vista and later.



Step 8: Validate Data Acquisition



Validating data acquisition involves calculating the hash value of the target media and comparing it with its forensic counterpart to ensure that the data is completely acquired



The unique number (hash value) is referred to as a **"digital fingerprint"**



As hash values are **unique**, if two files have the same hash value, they are 100% identical even if the files are named differently

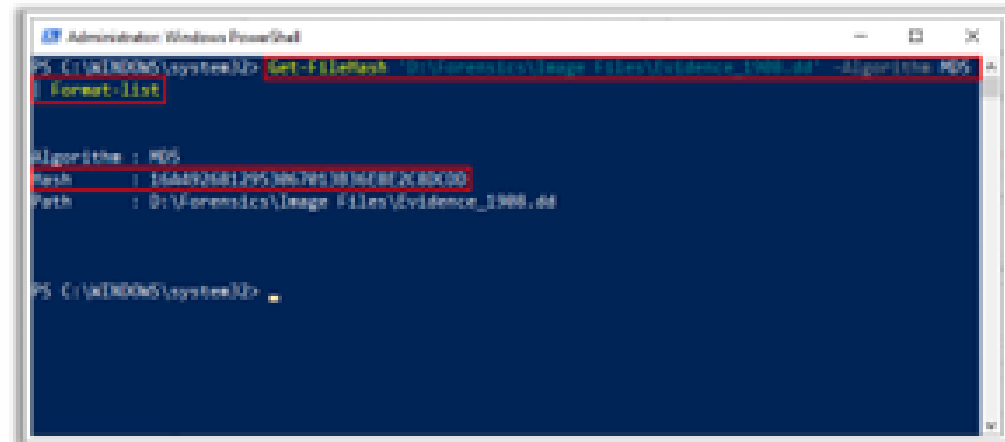


Utility algorithms that produce hash values include **CRC-32**, **MD5**, **SHA-1**, and **SHA-256**



Step 8: Validate Data Acquisition – Windows Validation Methods

- ☐ Windows computers come with **PowerShell** utility, which has the ability to run cmdlet
- ☐ The **Get-FileHash** cmdlet computes the hash value for an evidence file by using the specified hash algorithm
- ☐ This hash value is used throughout the investigation for validating the integrity of the evidence
- ☐ Investigators can also use commercial computer forensics programs, which have built-in validation features that can be used to validate the evidence files
- ☐ For instance:
 - ProDiscover's **.eve** files contain **metadata** in segmented files or acquisition files, including the hash value for the original media
 - When you load the image to ProDiscover, it compares the hash value of this image to the hash value of the original media
 - If the **hashes do not match**, the tool notifies that the image is corrupt, implying that the evidence cannot be considered reliable



```
Administrator: Windows PowerShell
PS C:\Windows\system32> Get-FileHash 'D:\forensics\image_files\Evidence_1988.dd' -Algorithm MD5
Format-List

Algorithm : MD5
Hash      : 16A8F068129538679538164C8F2C80000
Path      : D:\forensics\image_files\Evidence_1988.dd

PS C:\Windows\system32>
```

Note: In most computer forensics tools, raw format image files do not contain metadata. For raw acquisitions, therefore, a separate manual validation is recommended during analysis.



Module Summary



This module has discussed the data acquisition fundamentals



It has discussed the different types of data acquisition



It has also discussed in detail the data acquisition format



Finally, this module ended with a detailed discussion on data acquisition methodology



In the next module, we will discuss in detail on defeating anti-forensics techniques



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Defeating Anti-Forensic Techniques

SUBTOPIC 2



Module Objectives

- 1 Understanding the Anti-forensics Techniques
- 2 Understanding the Data Deletion and Recycle Bin Forensics
- 3 Overview of File Carving Techniques and Ways to Recover Evidence from Deleted Partitions
- 4 Understanding the Password Cracking/Bypassing Techniques
- 5 Understanding How to Detect Steganography, Hidden Data in File System Structures, and Trail Obfuscation
- 6 Understanding the Techniques of Artifact Wiping, Overwritten Data/ Metadata Detection, and Encryption
- 7 Overview of Anti-forensics Countermeasures and Anti-forensics Tools



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Module Flow

1

**Understand Anti-forensics
and its Techniques**

2

**Discuss Anti-forensics
Countermeasures**



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What is Anti-forensics?



Anti-forensics (also known as counter forensics) is a common term for a set of techniques aimed at **complicating or preventing a proper forensics investigation process**

Goals of Anti-forensics

- ✓ Interrupt and prevent information collection
- ✓ Make **difficult the investigator's task** of finding evidence
- ✓ Hide traces of crime or illegal activity
- ✓ Compromise the accuracy of a forensics report or testimony
- ✓ Delete evidence that an anti-forensics tool has been run



Anti-forensics Techniques

- | | | | |
|----------|---------------------------------------|-----------|-------------------------------------|
| 1 | Data/File Deletion | 7 | Overwriting Data/Metadata |
| 2 | Password Protection | 8 | Encryption |
| 3 | Steganography | 9 | Program Packers |
| 4 | Data Hiding in File System Structures | 10 | Minimizing Footprint |
| 5 | Trail Obfuscation | 11 | Exploiting Forensics Tool Bugs |
| 6 | Artifact Wiping | 12 | Detecting Forensics Tool Activities |



Anti-forensics Technique: Data/File Deletion



When a file is deleted from the hard drive, the **pointer to the file gets deleted** but the contents of file remain on the disk



In other words, the deleted files can be **recovered** from the hard disk **until** the sectors containing the contents of the file are **overwritten** with the new data



Data recovery Tools such as Autopsy, Recover My Files, EaseUS Data Recovery Wizard Pro, and R-Studio can be used for recovering deleted files/folders



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What Happens When a File is Deleted in Windows?



FAT File System

- ☐ The OS replaces the first letter of a deleted file name with a hex byte code: **E5h**
- ☐ **E5h** is a special tag that indicates that the file has been deleted
- ☐ The corresponding cluster of that file in FAT is marked as unused, although it will continue to contain the information until it is overwritten



NTFS File System

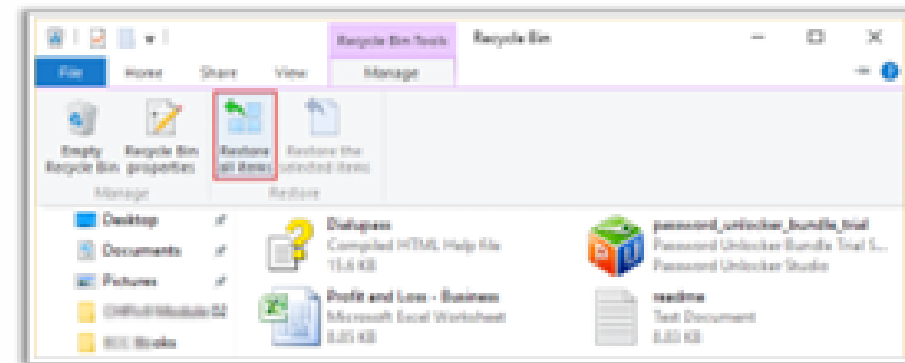
- ☐ When a user deletes a file, the OS just marks the file entry as unallocated but does not delete the actual file contents
- ☐ The clusters allocated to the deleted file are marked as free in the **\$BitMap** (\$BitMap file is a record of all used and unused clusters)
- ☐ The computer now notices those empty clusters and avails that space for storing a new file
- ☐ The deleted file can be recovered if the **space is not allocated** to any other file

Note: On a Windows system, performing normal **Delete** operation sends the files to the Recycle Bin. Whereas performing the **Shift+Delete** operation bypasses the Recycle Bin.



Recycle Bin in Windows

- 01** The Recycle Bin is a temporary storage place for deleted files
- 02** The file remains in the Recycle Bin until you empty the Recycle Bin or restore the file
- 03** Items can be restored to their original positions with the help of the **Restore all items** option of the Recycle Bin



Note: Deleting a file or folder from a network drive or from a USB drive may delete them permanently instead of being stored in the Recycle Bin



01

Recycle Bin in Windows (Cont'd)

When a file is deleted, the complete path of the file and its name is stored in a hidden file called **INFO2** (in Windows 98) in the Recycled folder. This information is used to restore the deleted files to their original locations.

02

Prior to Windows Vista, a file in the Recycle Bin was stored in its physical location and renamed using the syntax:

D<original drive letter of file><#>.<original extension>

- "D" denotes that a file has been deleted

03

Example:

De7.doc = (File is deleted from E: drive, it is the "eighth" file received by recycle bin, and is a "doc" file)



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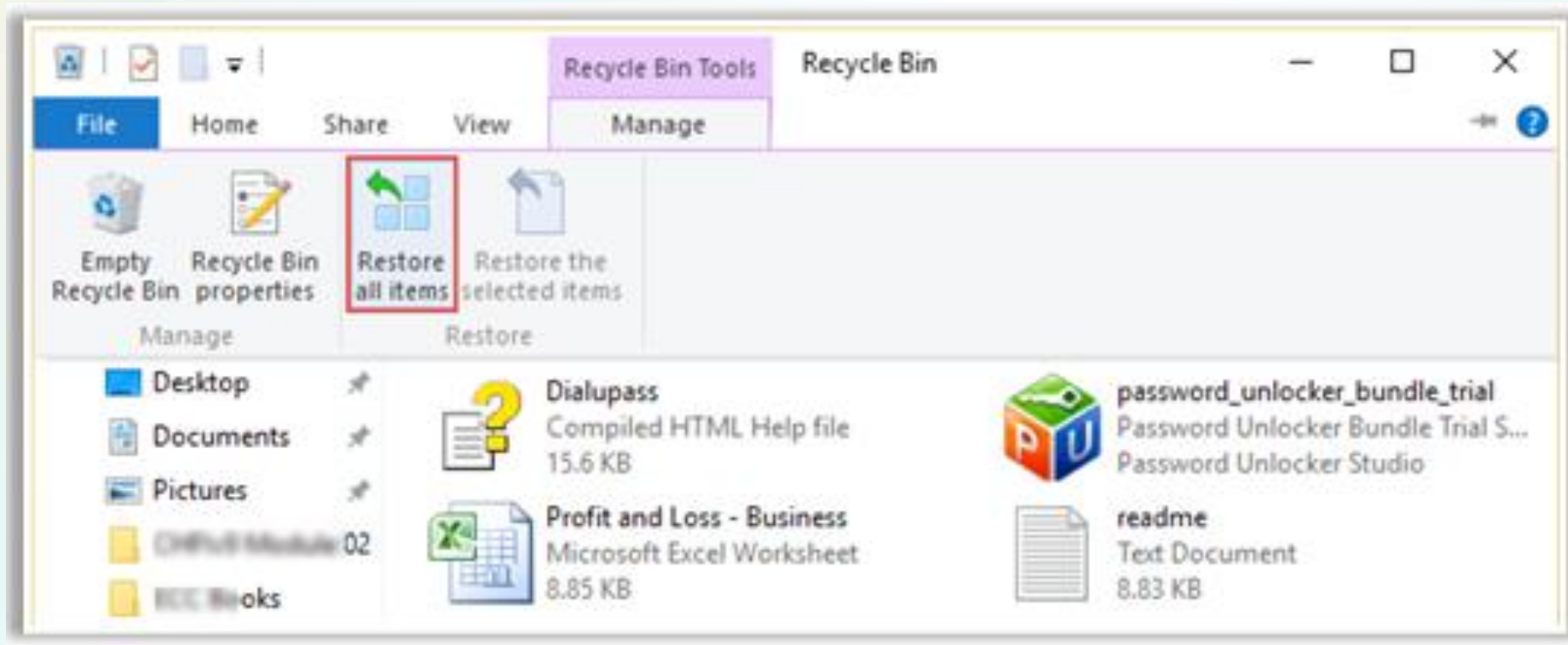


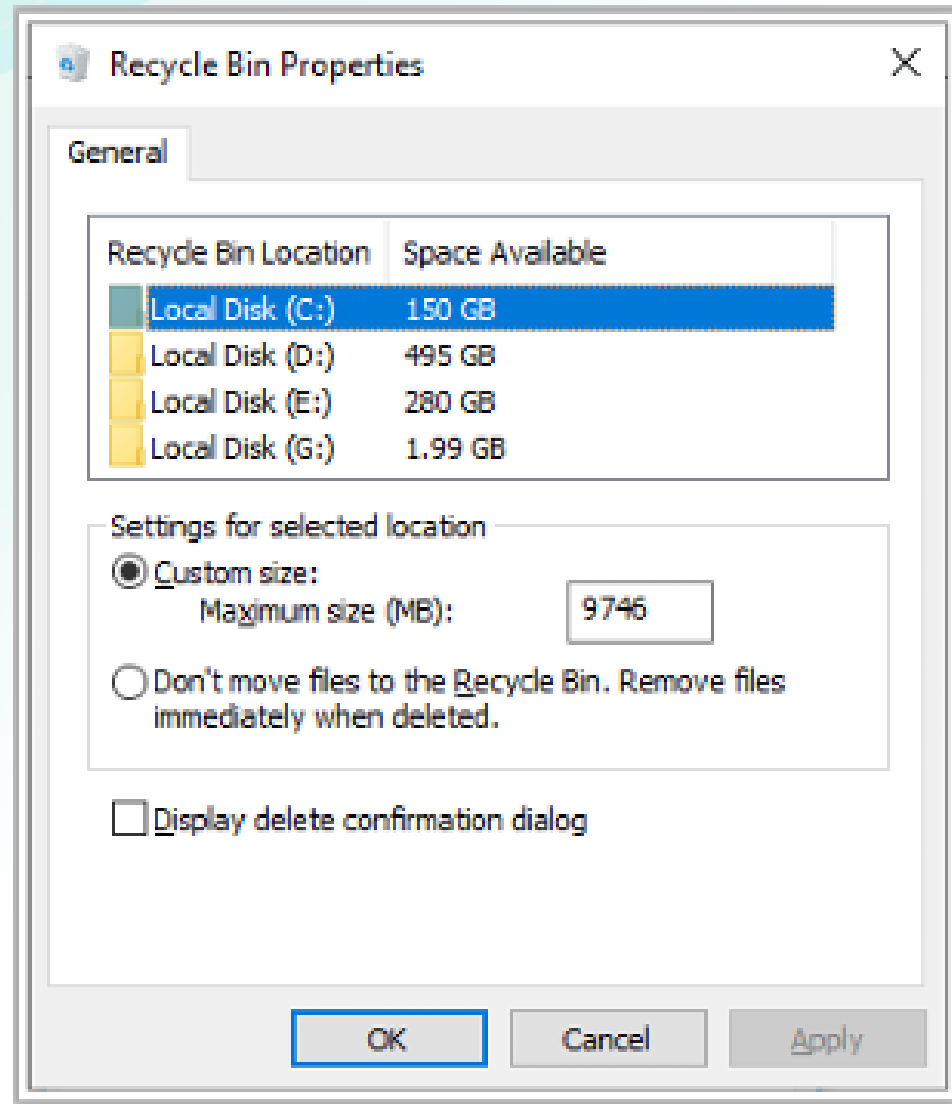
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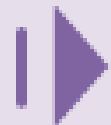




File Carving



- ❑ It is a technique to **recover files and fragments of files** from the hard disk in the absence of file system metadata



- ❑ In this technique, **file identification and extraction is based on certain characteristics** such as file header or footer rather than the file extension or metadata



- ❑ A file header is a **signature** (also known as a magic number), which is a constant numeric or text value that determines a file format

Example:

- A suspect may try to hide an image from being detected by investigators by changing the file extension from **.jpg** to **.dll**
- However, changing the file extension does not change the file header, and analysis tells the actual file format

Example:

- A file format is confirmed as **.jpg** if it shows **"JFIF"** in the file header and hex signature as "4A 46 49 46"

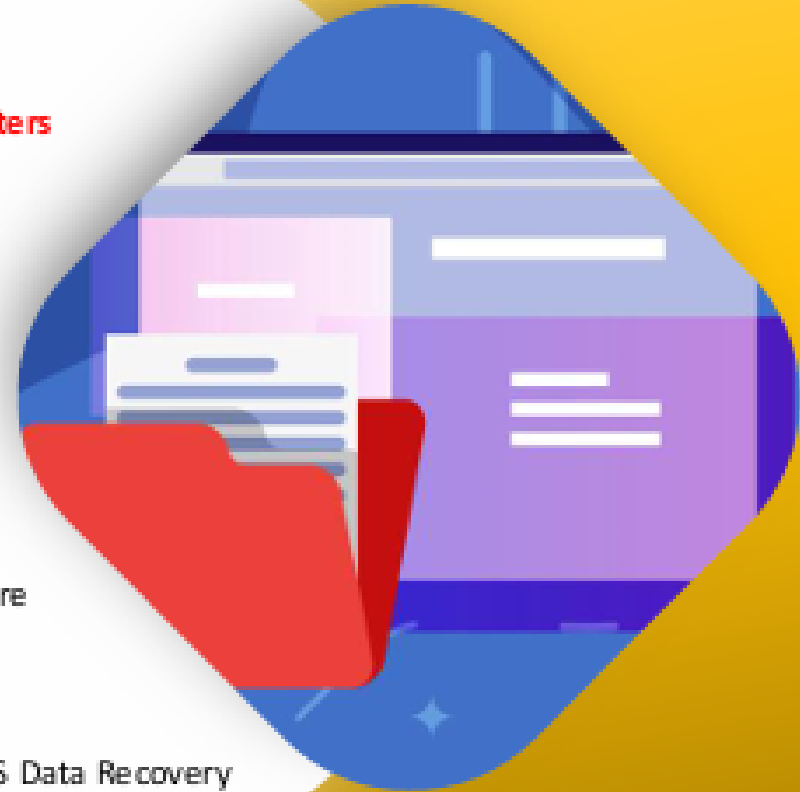


- ❑ Investigators can take a look at **file headers** to verify the file format using tools such as 010 Editor, CI Hex Viewer, Hexinator, Hex Editor Neo, Qjew, WinHex, etc.



File Carving on Windows

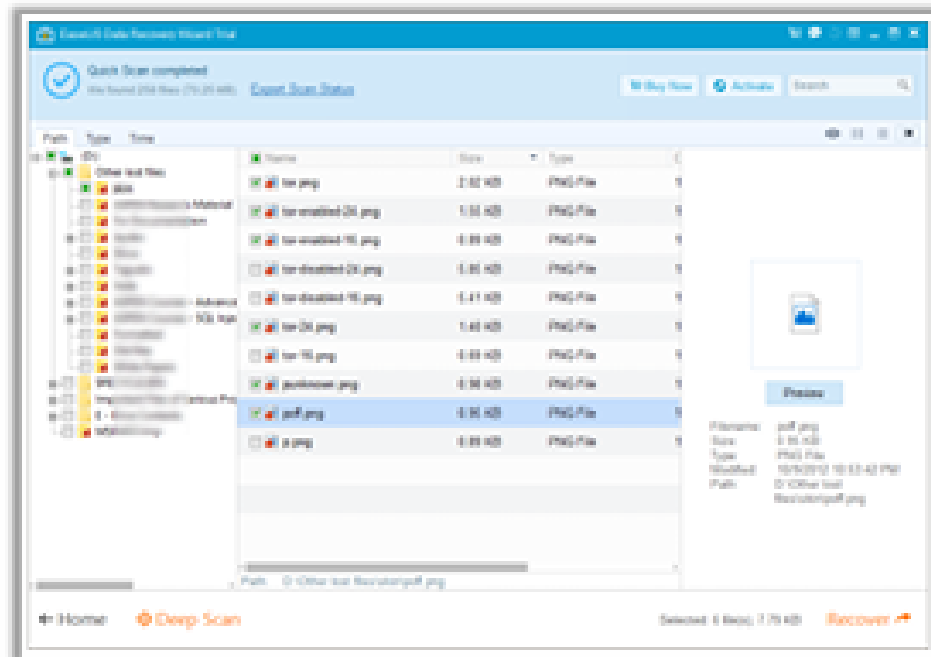
- ❑ Windows tracks its files/folders on a hard drive using the **pointers** that tells the system where the file begins and ends
- ❑ When a file is deleted from the hard drive, the pointer to the file gets deleted but the **contents of file remains on the disk**
- ❑ In other words, the deleted files can be recovered from the hard disk until the sectors containing the contents of the file are overwritten with new data
- ❑ **Data recovery Tools** such as Autopsy, Recover My Files, EaseUS Data Recovery Wizard Pro, R-Studio for Windows, etc., can be used for recovering deleted files/folders from Windows



File Recovery Tools: Windows

EaseUS Data Recovery Wizard Pro

Hard drive data recovery software to **recover lost data from PC**, laptop or other storage media due to deleting, formatting, partition loss, OS crash, virus attacks, etc.



IN THE U.S. ARMY, INDIVIDUAL CODE



Recover My Files:
<http://getdata.com>



Disk Digger
<https://diskdigger.org>



Handy Recovery
<http://www.handyrecovery.com>

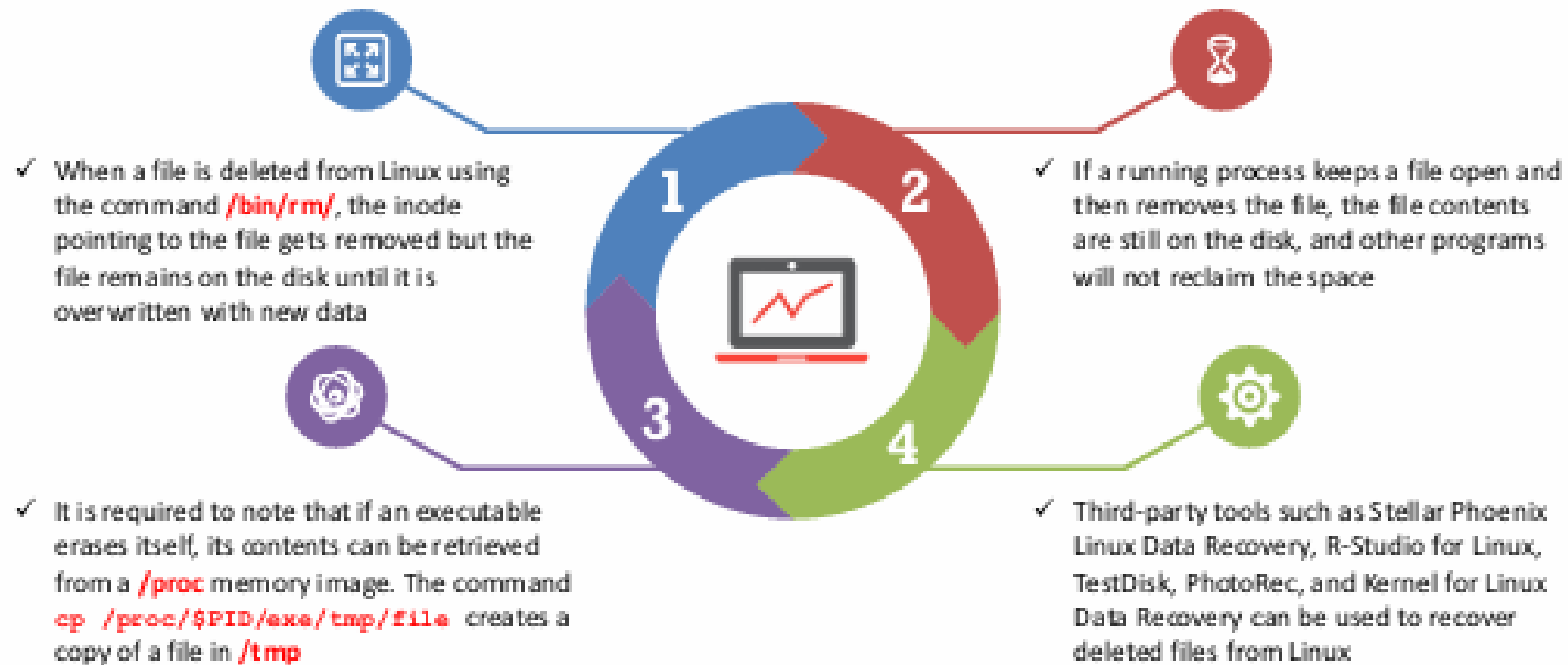


Quick Recovery
<http://www.recoveryanddata.com>



Stellar Phoenix Windows Data Recovery
<http://www.stellarinfo.com>

File Carving on Linux



SSD File Carving on Linux File System



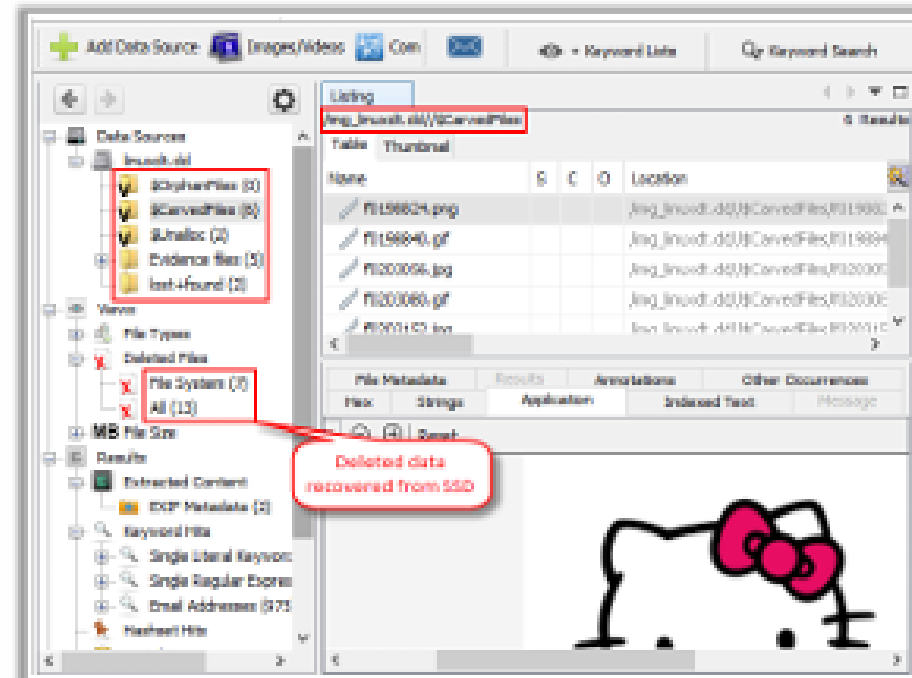
Forensic workstation used: **Windows 10**

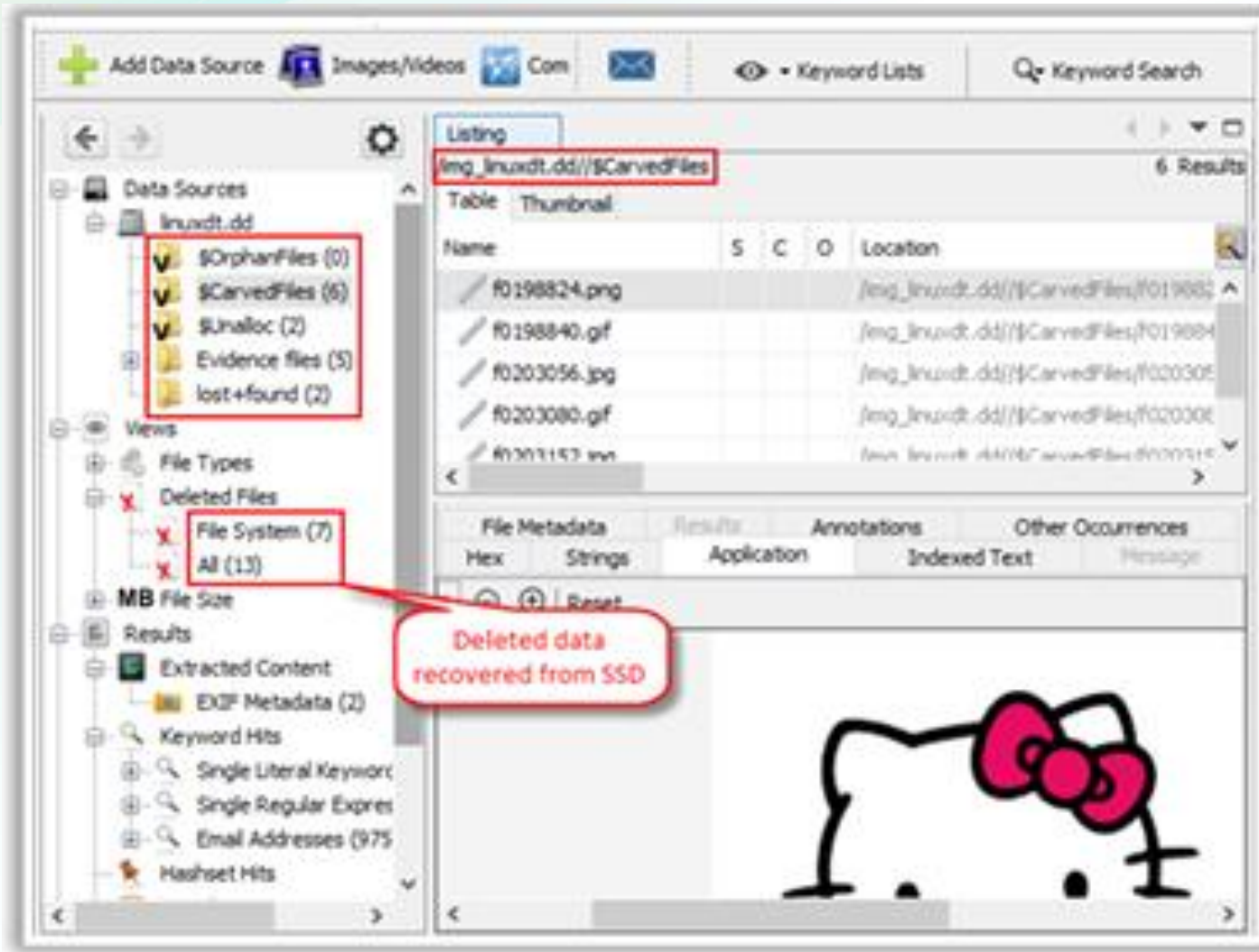


The forensically acquired image from **TRIM disabled SSD** should be examined using file carving tools such as Autopsy, R-Studio, etc.



In Autopsy, the carved data from the forensic evidence file is displayed under the appropriate data source with heading "**\$CarvedFiles**"





Recovering Deleted Partitions



The MBR partition table **contains the records** of the primary and extended partitions of a disk



When a **partition is deleted** from a disk, the entries with respect to deleted partition are removed by the computer from the MBR partition table



Investigators use **tools** such as R-Studio and EaseUS Data Recovery Wizard to scan the disk for lost partitions and recover them



These automated tools perform full disk scan, looks for **deleted partition information** and reconstruct the partition table entry for deleted partition



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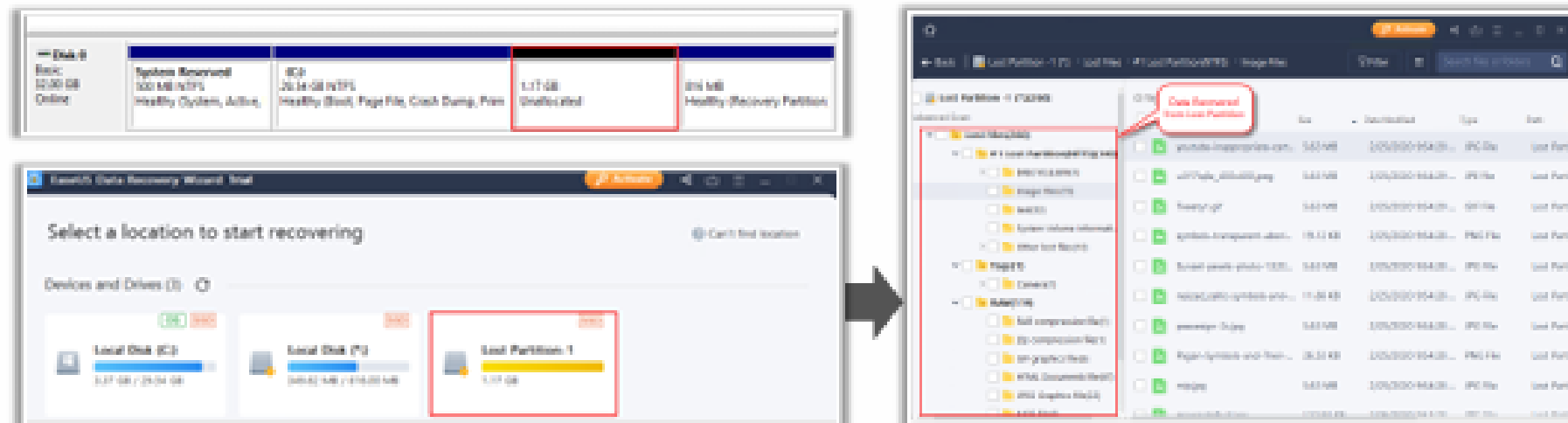
Recovering Deleted Partitions: Using EaseUS Data Recovery Wizard



When an **Unallocated partition** is found during forensic investigation, the investigator use automated tools such as EaseUS Data Recovery Wizard to discover lost partition and recover the data from it



EaseUS Data Recovery Wizard tool **recovers data from the FAT and NTFS** based file system partitions





Anti- forensics Technique: Password Protection



While conducting forensic investigation on suspect's computer, **accessing password protected resources** is one of the challenges faced by the investigator



In such cases, investigators can use password cracking tools such as ophcrack and RainbowCrack to **circumvent** the password protection



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Password Types



Cleartext Passwords

Cleartext passwords are transmitted or stored on media **without any encryption**



Obfuscated Passwords

Obfuscated **passwords are encrypted** using an algorithm and can be decrypted by applying a reverse algorithm



Password Hashes

Password hashes are **signatures** of the original password, generated using a one-way algorithm. Passwords hashed using hash algorithms (MD5, SHA, etc.) are not reversible



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Password Cracking Techniques

Dictionary Attack

- A **dictionary file is loaded** into the cracking application that runs against user accounts

Brute Forcing Attacks

- The program **tries every combination** of characters until the password is broken

Rule-based Attack

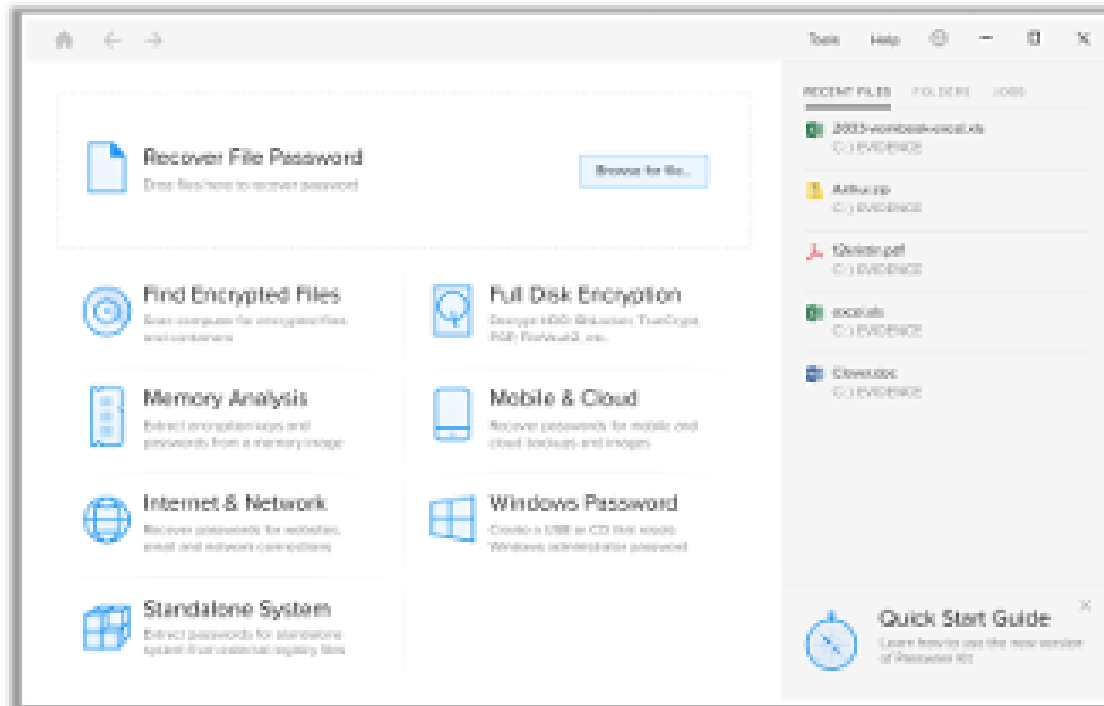
- This attack is used when some **information** about the password **is known**



Password Cracking Tools

Passware Kit Forensic

- ☐ A complete **encrypted electronic evidence discovery** solution that reports and decrypts all password-protected items on a computer



<https://leewarepassware.com>



L0phtCrack

<https://www.l0phtcrack.com>



oph crack

<https://ophcrack.sourceforge.io>



Cain & Abel

<https://www.wisid.it>



RainbowCrack

<https://project-rainbowcrack.com>



Offline NT Password & Registry Editor

<https://pogostick.net>



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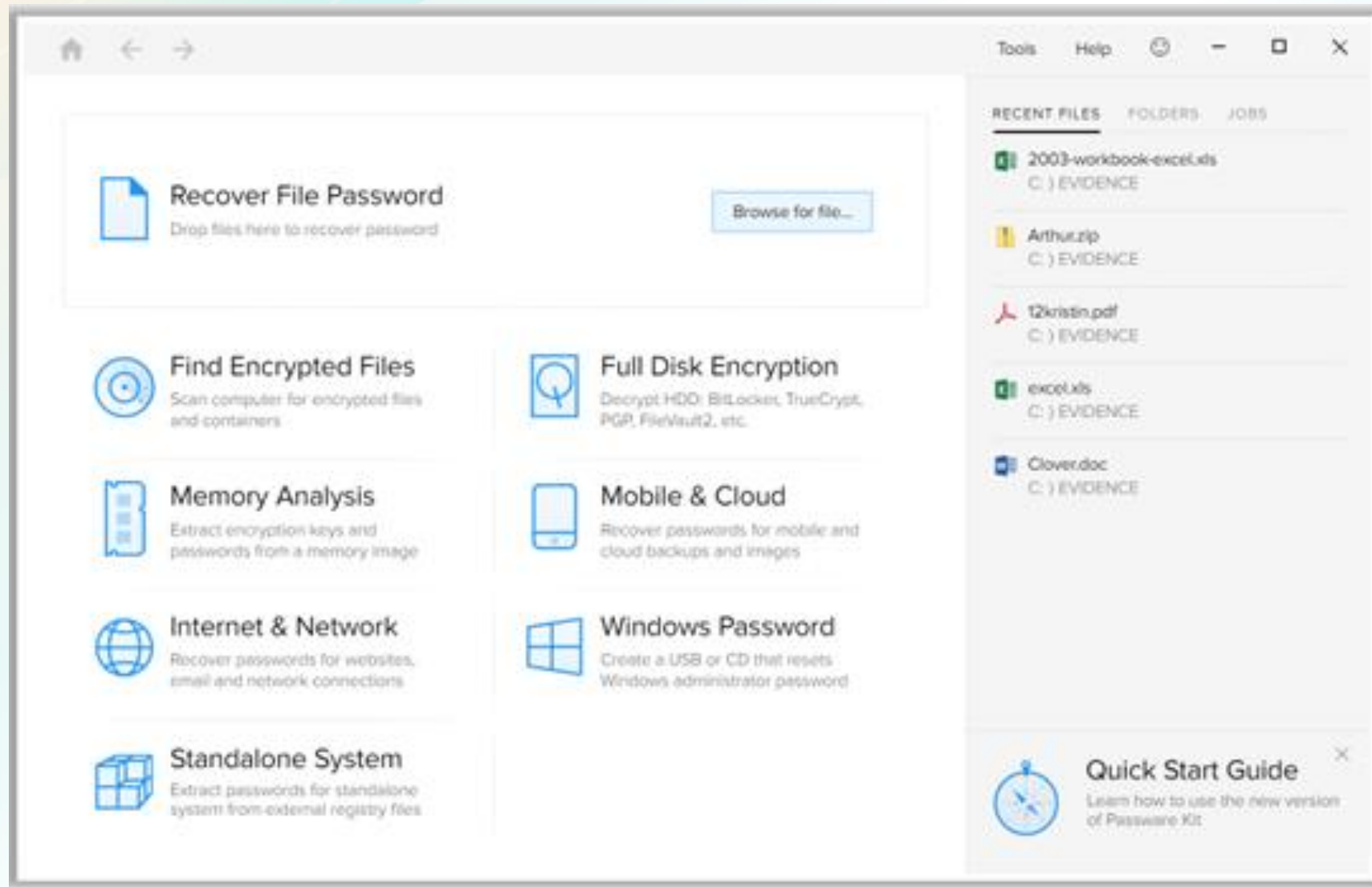


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Anti-forensics Technique: Steganography



Steganography is a technique of **hiding a secret message** within an ordinary message and **extracting it at the destination** to maintain confidentiality of data



Attacker can use steganography to hide messages such as **a list of the compromised servers**, source code for the hacking tool, and plans for future attacks



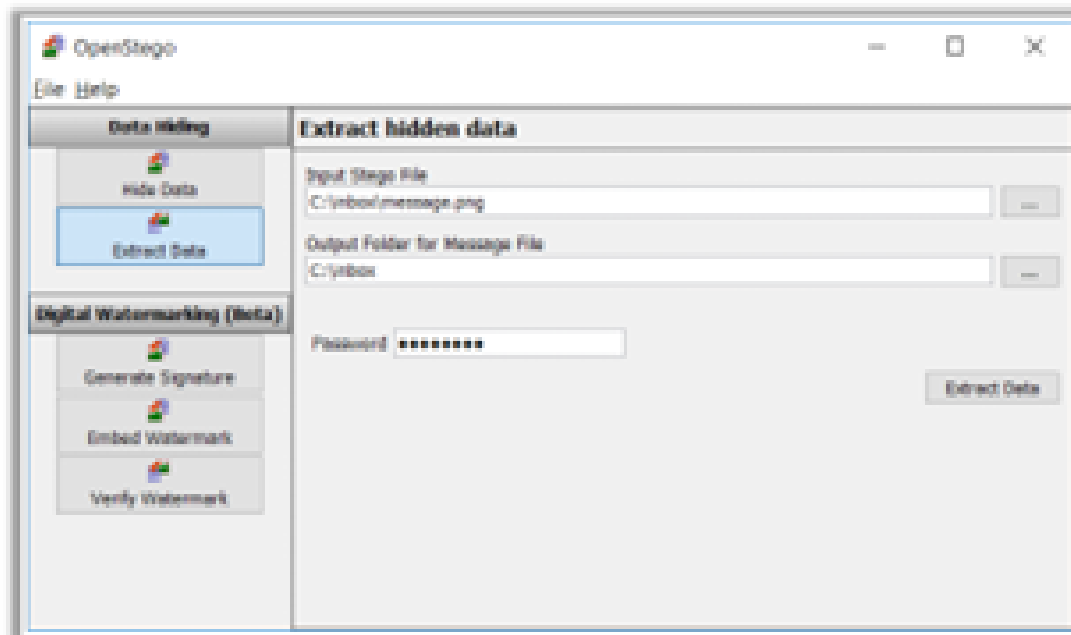
The forensic investigator examines the **Stego-Object** to extract the hidden information



Steganography Detection Tools

OpenStego

- ❑ OpenStego provides two main functionalities: **data hiding** and **watermarking**. It allows investigators to extract hidden data for analysis.



<https://www.seopenstego.com/>

StegSpy

- ❑ StegSpy identifies a "**steganized**" file and detects steganography and the program used to hide the message



<http://www.spy-hunter.com/>



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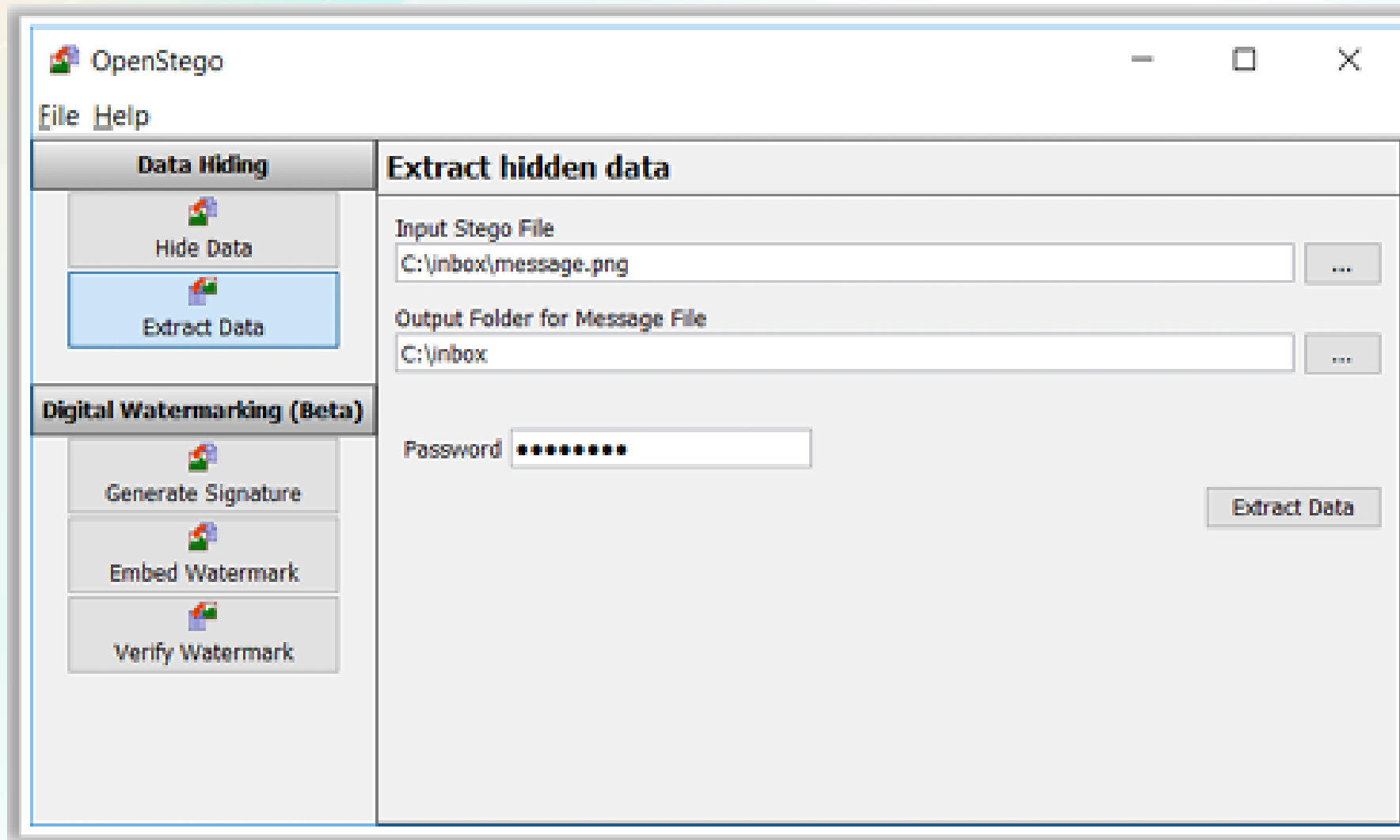


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Anti-forensics Technique: Alternate Data Streams



Attackers use **Alternate Data Streams (ADS)** to **hide data** in Windows NTFS and cannot be revealed through command line or Windows Explorer



ADS allows **attacker to hide** any number of streams into one single file without modifying the file size, functionality, etc., except the file date



However, the file date can be modified using anti-forensics tools like TimeStomp



In some cases, these hidden ADS can be used to **remotely exploit** a web server



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Anti-forensics Technique: Trail Obfuscation

- ❑ The purpose of trail obfuscation is to **confuse** and **mislead** the forensics investigation process
- ❑ Attackers **mislead investigators** via log tampering, false e-mail header generation, timestamp modification, and various file headers' modification

Some of the techniques attackers
use for data/trail obfuscation:

- ✓ Log cleaners
- ✓ Spoofing
- ✓ Misinformation
- ✓ Zombie accounts
- ✓ Trojan commands

Traffic content obfuscation can be attained by means of VPNs and SSH tunneling



Anti-forensics Technique: Trail Obfuscation (Cont'd)



Timestamp is one of the most widely used trail obfuscation tools that allow **deletion** or **modification** of **timestamp-related** information on files. Procedure to defeat this technique is covered in "Detecting Overwritten Data/Metadata" section.



```
root@kali: ~  
File Edit View Search Terminal Help  
meterpreter > timestamp secret.txt -v  
Modified      : 2014-03-07 12:56:07 +0530  
Accessed      : 2014-03-07 12:50:51 +0530  
Created       : 2014-03-07 12:45:51 +0530  
Entry Modified: 2014-03-07 12:56:07 +0530  
meterpreter >
```

```
root@kali: ~  
File Edit View Search Terminal Help  
meterpreter > timestamp secret.txt -v  
Modified      : 2014-03-07 12:56:07 +0530  
Accessed      : 2014-03-07 12:50:51 +0530  
Created       : 2014-03-07 12:45:51 +0530  
Entry Modified: 2014-03-07 12:56:07 +0530  
meterpreter > timestamp secret.txt -m "06/15/2012 12:57:37"  
[*] Setting specific MACE attributes on secret.txt  
meterpreter >
```

```
root@kali: ~  
File Edit View Search Terminal Help  
Accessed      : 2014-03-07 12:50:51 +0530  
Created       : 2014-03-07 12:45:51 +0530  
Entry Modified: 2014-03-07 12:56:07 +0530  
meterpreter > timestamp secret.txt -m "06/15/2012 12:57:37"  
[*] Setting specific MACE attributes on secret.txt  
meterpreter > timestamp secret.txt -a "06/15/2012 12:55:05"  
[*] Setting specific MACE attributes on secret.txt  
meterpreter >
```

```
root@kali: ~  
File Edit View Search Terminal Help  
Entry Modified: 2014-03-07 12:56:07 +0530  
meterpreter > timestamp secret.txt -m "06/15/2012 12:57:37"  
[*] Setting specific MACE attributes on secret.txt  
meterpreter > timestamp secret.txt -a "06/15/2012 12:55:05"  
[*] Setting specific MACE attributes on secret.txt  
meterpreter > timestamp secret.txt -c "06/12/2012 12:50:22"  
[*] Setting specific MACE attributes on secret.txt  
meterpreter >
```

```
root@kali: ~  
File Edit View Search Terminal Help  
[*] Setting specific MACE attributes on secret.txt  
meterpreter > timestamp secret.txt -a "06/15/2012 12:55:05"  
[*] Setting specific MACE attributes on secret.txt  
meterpreter > timestamp secret.txt -c "06/12/2012 12:50:22"  
[*] Setting specific MACE attributes on secret.txt  
meterpreter > timestamp secret.txt -e "06/15/2012 13:59:40"  
[*] Setting specific MACE attributes on secret.txt  
meterpreter >
```

```
root@kali: ~  
File Edit View Search Terminal Help  
meterpreter > timestamp secret.txt -v  
Modified      : 2012-06-15 13:57:37 +0530  
Accessed      : 2012-06-15 13:55:05 +0530  
Created       : 2012-06-12 13:50:22 +0530  
Entry Modified: 2012-06-15 14:59:48 +0530  
meterpreter >
```

```
root@kali: ~  
File Edit View Search Terminal Help  
meterpreter > timestamp secret.txt -v  
Modified      : 2014-03-07 12:56:07 +0530  
Accessed      : 2014-03-07 12:50:51 +0530  
Created       : 2014-03-07 12:45:51 +0530  
Entry Modified: 2014-03-07 12:56:07 +0530  
meterpreter >
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[*] Setting specific MACE attributes on secret.txt  
meterpreter >
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[*] Setting specific MACE attributes on secret.txt  
meterpreter > timestamp secret.txt -a "06/15/2012 12:55:05"  
[*] Setting specific MACE attributes on secret.txt  
meterpreter > timestamp secret.txt -c "06/12/2012 12:50:22"  
[*] Setting specific MACE attributes on secret.txt  
meterpreter >
```

```
root@kali: ~  
File Edit View Search Terminal Help  
[*] Setting specific MACE attributes on secret.txt  
meterpreter > timestamp secret.txt -a "06/15/2012 12:55:05"  
[*] Setting specific MACE attributes on secret.txt  
meterpreter > timestamp secret.txt -c "06/12/2012 12:50:22"  
[*] Setting specific MACE attributes on secret.txt  
meterpreter > timestamp secret.txt -e "06/15/2012 13:59:48"  
[*] Setting specific MACE attributes on secret.txt  
meterpreter >
```

```
root@kali: ~  
File Edit View Search Terminal Help  
meterpreter > timestamp secret.txt -v  
Modified      : 2012-06-15 13:57:37 +0530  
Accessed      : 2012-06-15 13:55:05 +0530  
Created       : 2012-06-12 13:50:22 +0530  
Entry Modified: 2012-06-15 14:59:48 +0530  
meterpreter >
```

Timestamp

Anti-forensics Technique: Artifact Wiping



Artifact wiping involves various methods aimed at **permanent deletion** of particular files or entire file systems

Artifact wiping methods:



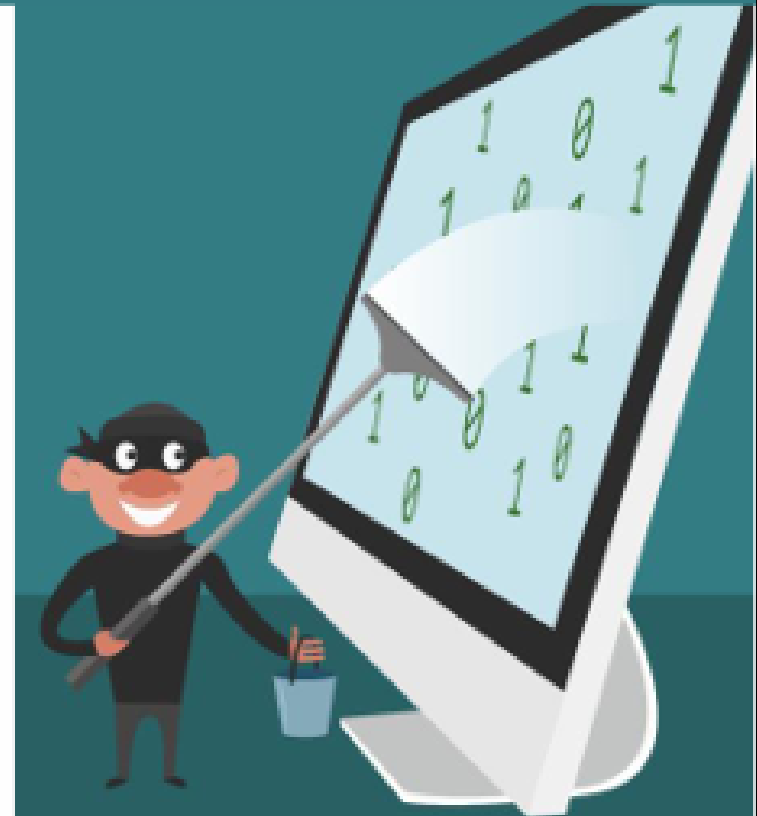
1. Disk Wiping Utilities

- Disk wiping involves **erasing data** from the disk by **deleting its links to memory blocks** and overwriting the memory contents
- Some of the commonly used disk wiping utilities include BCWipe Total WipeOut, CyberScrub cyberCide, DriveScrubber, ShredIt, etc.



2. File Wiping Utilities

- Deletes individual files and file table entries from an OS
- Some of the commonly used **file wiping** utilities include BCWipe, R-Wipe & Clean, CyberScrub Privacy Suite, etc.



Anti-forensics Technique: Artifact Wiping (Cont'd)

3. Disk Degaussing/Destruction Techniques

- ☐ Disk degaussing is a process by which a strong **magnetic field** is applied to storage device, resulting in an entirely clean device of any previously stored data
- ☐ NIST recommends a variety of methods to accomplish **physical destruction of the digital media**, which includes disintegration, incineration, pulverizing, shredding, and melting
- ☐ Intruders use disk degaussing/destruction techniques to **make the evidentiary data unavailable** to forensics investigators

4. Disk Formatting

- ☐ Formatting of a hard drive **does not erase** the data present on the disk but **wipes its address tables** and unlinks all the files in the file system
- ☐ Later, a new file tree is set up to use with OS
- ☐ After formatting a hard disk, the forensic investigator **can recover data** from a formatted drive

Note: It is difficult to retrieve data that has been wiped out using these techniques. Hence, it is recommended that the organizations take a **back up of the data** on regular basis.



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Anti-forensics Technique: Overwriting Data/Metadata



- ❑ Perpetrators use different techniques to **overwrite data or metadata** (or both) on a storage media thereby posing a challenge to the investigators while performing data recovery
- ❑ Overwriting programs (disk sanitizers) work in three modes:
 - ✓ Overwrite entire media
 - ✓ Overwrite individual files
 - ✓ Overwrite deleted files on the media

Overwriting Metadata

- ❑ Investigators use metadata to **create a timeline** of the perpetrator's actions by arranging all the computer's timestamps in a sequence
- ❑ Although attackers can use tools to wipe the contents of media, that action itself might draw the attention of investigators. Therefore, attackers cover their tracks by overwriting the metadata (i.e. access times), rendering the construction of timeline difficult.
- ❑ **Ex:** Timestomp (part of the Metasploit Framework) is used to change MACE (Modified-Accessed-Created-Entry) attributes of the file



Anti-forensics Technique: Encryption



- ❑ Data encryption is one of the commonly used techniques to **defeat forensics investigation process**



- ❑ Intruders use strong encryption algorithms to encrypt data of investigative value, which renders it virtually unreadable without the **designated key**



- ❑ Additionally, most encryption programs are capable of performing additional functions, including use of a key file, **full-volume encryption**, and plausible deniability; that makes the investigator's job more difficult



- ❑ Cryptanalysis can be used to decrypt encrypted data
 - **Example:** CrypTool

Module Flow

1

**Understand Anti-forensics
and its Techniques**

2

**Discuss Anti-forensics
Countermeasures**



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Anti-forensics Countermeasures



- 1 Train and educate the forensic investigators about anti-forensics
- 2 Validate the **results of examination** using multiple tools
- 3 Impose **strict laws** against illegal use of anti-forensics tools
- 4 Understand the anti-forensic techniques and their weaknesses
- 5 Use latest and updated CFTs and test them for **vulnerabilities**

Anti-forensics Tools



Steganography Studio

<http://stegstudio.sourceforge.net>



CryptaPix

<https://www.briggssoft.com>



GiliSoft File Lock Pro

<http://gilisoft.com>



wbStego

<https://wbstego.wbailer.com>



Data Stash

<https://www.skyjuicesoftware.com>



OmniHide PRO

<https://omnihide.com>



Masker

<http://softpub.weebly.com>



DeepSound

<http://pinsoft.net>



DBAN

<https://dban.org>



east-tec InvisibleSecrets

<https://www.east-tec.com>



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Module Summary



- ➔ This module has discussed the anti-forensics techniques
- ➔ It has discussed the data deletion and Recycle Bin forensics
- ➔ It has also discussed in detail the file carving techniques and ways to recover evidence from deleted partitions
- ➔ This module has also discussed the password cracking/bypassing techniques
- ➔ It has also discussed how to detect steganography, hidden data in file system structures, and trail obfuscation
- ➔ It has also discussed the techniques of artifact wiping, overwritten data/metadata detection, and encryption
- ➔ Finally, this module ended with a detailed discussion on the anti-forensics countermeasures and anti-forensics tools
- ➔ In the next module, we will discuss in detail on Windows forensics

REFERENCES

- **Digital Forensics Essentials – 2021 by EC-Council**